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March 4, 1959

MEMO TO: Mr. D. B. Miller

SUBJECT: Technical Paper Activity in the 7030 Area

Reference: Memo by D. G. Knight, of February 27.

It might be well to summarize the current publication activities in the 7030 area.

The following systems planning papers are at various stages of active preparation:

1. "Processing Data in Bits and Pieces" by F. P. Brooks, Jr., G. A. Blaauw, and W. Buchholz. Completed. Accepted for presentation at the ICIP, Paris, June 1959. Distributed as internal report TR 00.01000.674. Submitted for publication to the IRE Transactions on Electronic Computers.

2. "Indexing and Control Word Techniques" by G. A. Blaauw, a considerably expanded version of a paper presented under a similar title at the December 1958 EJCC. Completed. Submitted for publication to the IBM Journal.

3. "Fingers or Fists?" by W. Buchholz. Completed. Submitted for publication to the IBM Journal.

4. "The Execute Operations - A New Type of Instruction Sequencing" by F. P. Brooks, Jr. In preparation. A communication to be submitted to the IBM Journal.

5. "A Multiplexed Input-Output System for On-Line Computer Operation" by W. Buchholz. In preparation.

The following systems planning papers are being considered for later this year:

6. A console paper by F. P. Brooks, Jr.

7. A paper on instruction exceptions by C. A. Scalzi.

8. A floating point paper by D. W. Sweeney.

9. A paper on the Virtual Memory by J. Cocke.

10. A paper on timing simulation by H. G. Kolsky.

11. Multiple Specialized Computers by S. W. Dunwell.

The following "hardware" papers are being considered by the design groups for preparation later this year.

12. Systems Organization of the Sigma Computer by E. Bloch and E. D. Foss.

13. A Multiple Divisor Divide Scheme for High Speed Binary Arithmetic by J. R. Stewart.

14. Multiplexed Multiply Schemes for High Speed Computers by O. L. MacSorley and F. R. Bielawa.

15. The Series-Parallel Arithmetic Unit of Sigma Computer by E. Bloch.

16. The Sequential Operation on the Systems Level (Pipelining) by J. A. Hipp and S. G. Tucker.

17. A Bus System for Shared Diverse Speed Asynchronous Memories by R. T. Blosk and L. O. Ulfsparre.

These lists are, of course, subject to expansion. We are also thinking about collecting some of these papers and republishing them as a book.

My conclusion is that there is no decline in this particular area. While pressure of other work always tends to slow down publication effort, it is realized that this is the year when publication work on the 7030 system should be at its peak if we are to get the benefits of timely publication. I think that the incentive is high enough that we will maintain a good level of activity until 7030 documentation is essentially complete.

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cc: E. Bloch
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